# Migration Costs and Higher Education: Experimental Evidence from India

Merve Demirel
IIES, Stockholm University

November 1, 2026

# Puzzle: Low migration rates despite large spatial opportunity gaps

- Large spatial gaps in economic opportunity coexist with low internal migration rates. Why?
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**Research question: What keeps students from moving for better educational opportunities?**

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- Quantify their perceived costs and benefits of education and labor market outcomes
- Randomize access to mentorship to learn how beliefs and migration choices respond to information

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$\Rightarrow$ Findings consistent with **real non-pecuniary costs** of migration as a key barrier to mobility

# This paper

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  Discrete choice experiment and data on destination-specific expectations

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- **Test** whether beliefs and migration choices are malleable
  RCT with 1,682 students: randomize offer of a phone mentorship with a migrant peer
  - → Applications away from home **+4.0 pp**, migration **+2.7 pp** (index **+0.16 SD**)
  - → Treatment effects are concentrated in **short-distance moves by men**
  - → Women do not respond to the intervention, and their beliefs **worsen**

# Contributions to the literature

- Migration costs are large, with limited information on what constitutes them
  (Sjaastad, 1962; Kennan et al., 2011; Baláž et al., 2016; Bryan et al., 2019; Fu et al., 2022; Koşar et al., 2022; Sahai et al., 2022; Lagakos et al., 2023; Fabre, 2024; Koenen et al., 2024; Ishimaru, 2025; Batista et al., 2025; Ghose, 2026)

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  → Subjective - and broadly accurate - beliefs limit mobility beyond day-to-day commuting

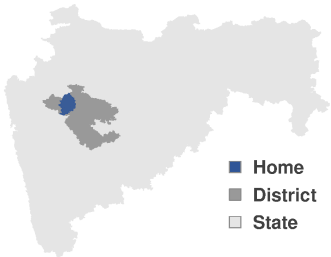
# Today's talk

1. Setting and data
2. Conceptual framework
3. Experiment I: Design
4. Findings I: The non-pecuniary costs of migration
5. Experiment II: Design
6. Findings II: Effects of mentorship
7. Conclusion

# Setting and data

# Educational migration in Maharashtra

**An applicant's geography**



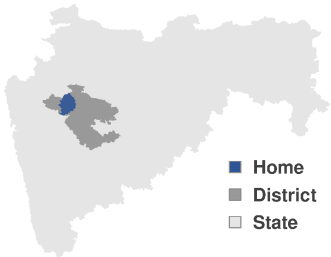

Source: administrative admissions records of the State Common Entrance Test Cell, Government of Maharashtra.

25th percentile 75th percentile Quality

# Educational migration in Maharashtra

- High **observational** returns: increases in number and quality of college options, income

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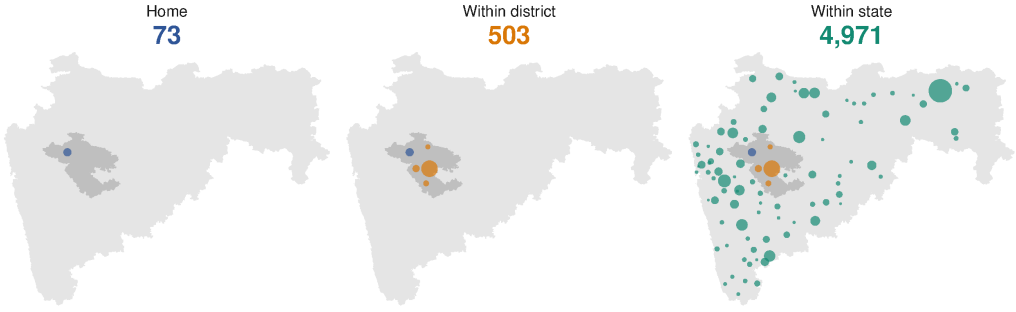

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**Number of accessible spots per qualifying candidate, median engineering applicant**

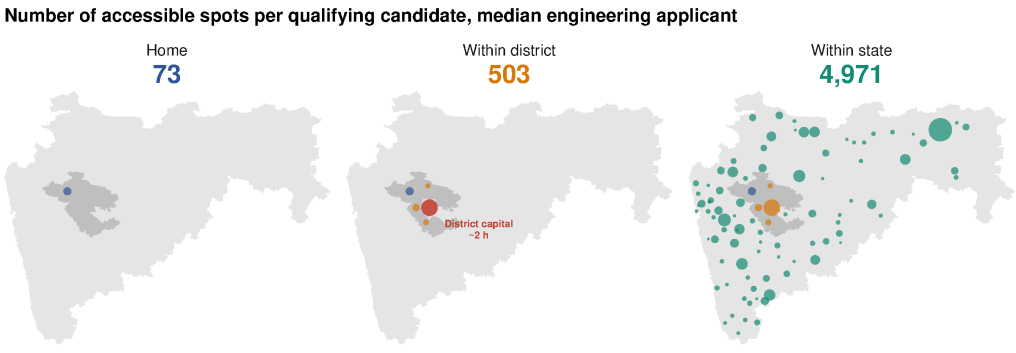



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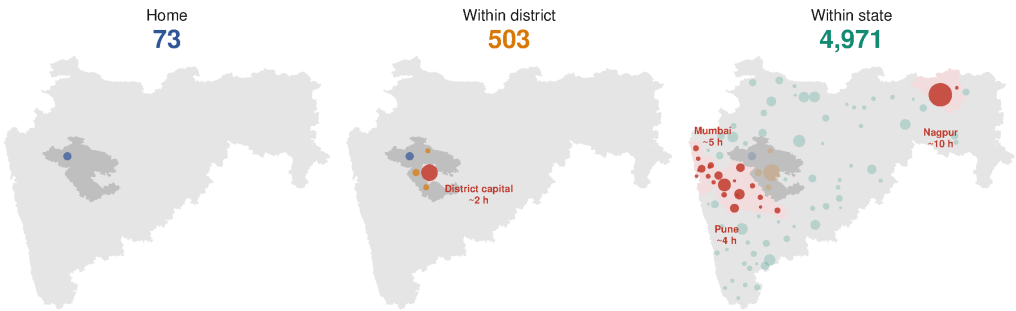

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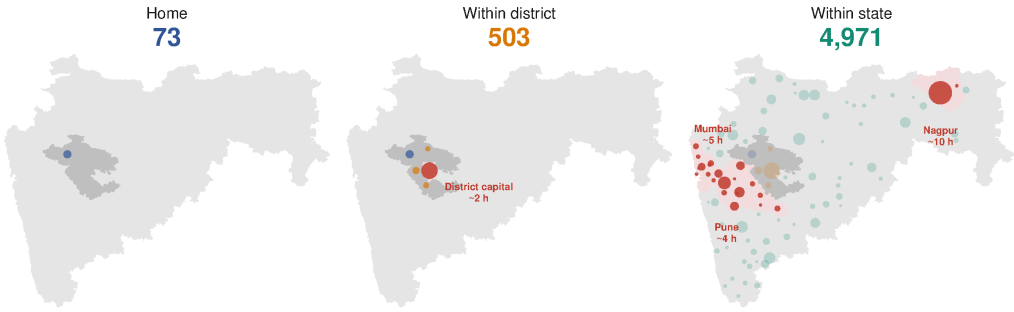

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# Educational migration in Maharashtra

- High **observational** returns: increases in number and quality of college options, income
- Yet moving for college is not common: 1.7% of college-age population move (Census)

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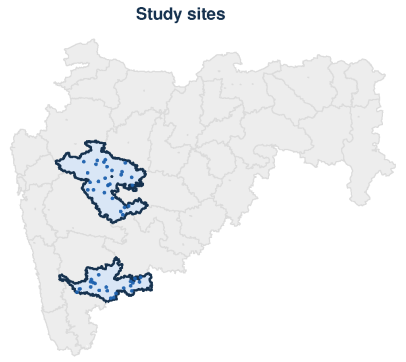



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# Study sites and sample

- Two districts in Maharashtra: Ahilyanagar and Sangli
  - 1,682 final-year secondary students in 52 schools
  - All *aiming* to study further and work
- Baseline survey serves as application to the program
  1. To make the survey incentive compatible
  2. To screen out uninterested students
- Educational migration **feasible** on average:
  - Academically: *expect* to score at median
  - Financially: comparable budget to a representative student in Pune (Mangal, 2024)



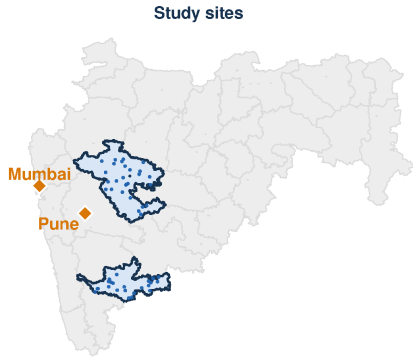
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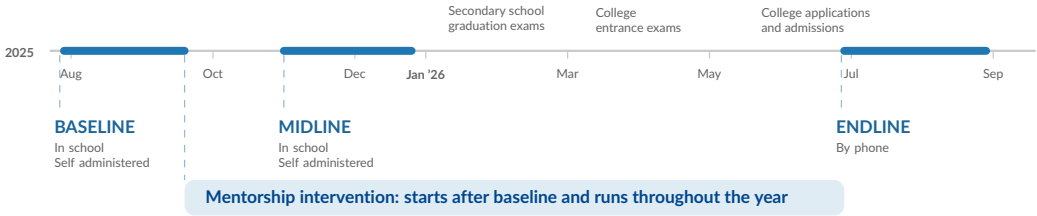
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- Mentors located in Mumbai and Pune
  - .. *more on the mentorship program later*



# Data and timeline




- **Student surveys** three rounds
- **Program records** mentorship call records, mentor reports and application data

- **Findings I** data from baseline and control group midline
- **Findings II** data from endline

# Conceptual framework

# Students' educational migration decision

Consider the utility of individual $i$ studying in destination $d$

$$U_i\Big(\underbrace{v_d}_{\substack{\text{pecuniary}\\ \text{outcomes}}}, \quad \underbrace{n_d}_{\substack{\text{non-pecuniary}\\ \text{outcomes}}}\Big), \qquad d \in D_i$$

$D_i$: home and the away destinations student $i$ considers; $U_i$ increasing in both arguments

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A student migrates if expected utility from studying in $d$ is larger than from studying at home

$$\Delta U_{id} \equiv E_i\big[U_i(v_d, n_d)\big] - E_i\big[U_i(v_h, n_h)\big] > 0$$

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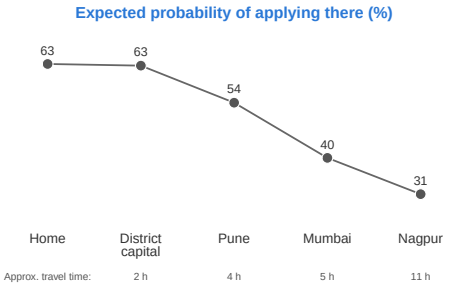
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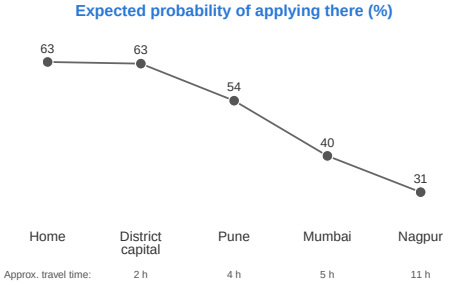
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# Students' stated choice probability falls despite high perceived returns




By gender

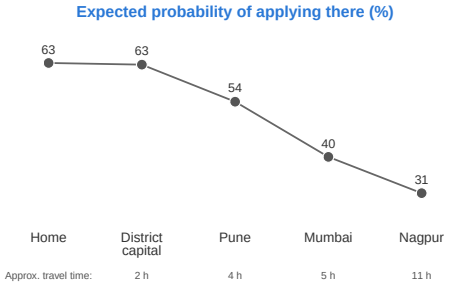
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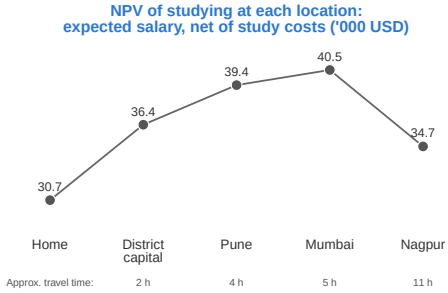

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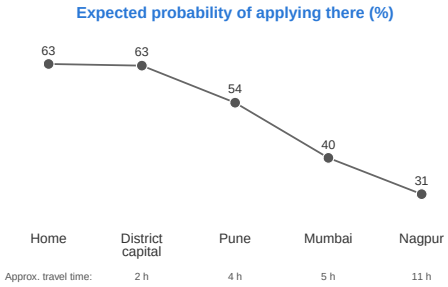

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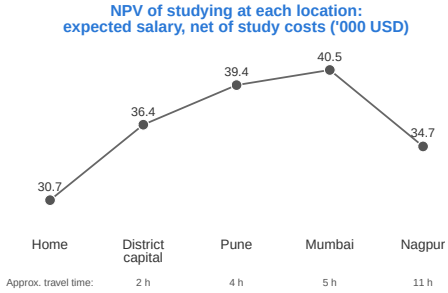

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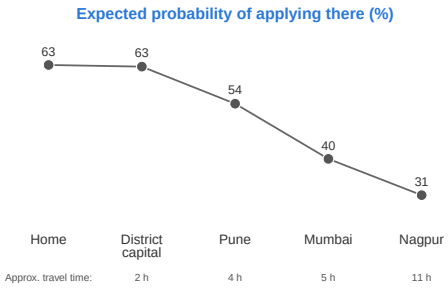



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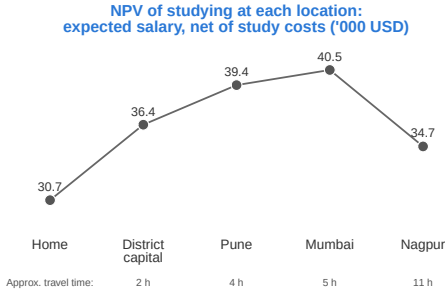
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By gender

# Students' educational migration decision

**Experiment I:** What is the role of non-pecuniary considerations?

1. How large is $\Delta NonPec_{id}$?
   Pecuniary and non-pecuniary outcomes at destinations are correlated $\Rightarrow$ vary them independently

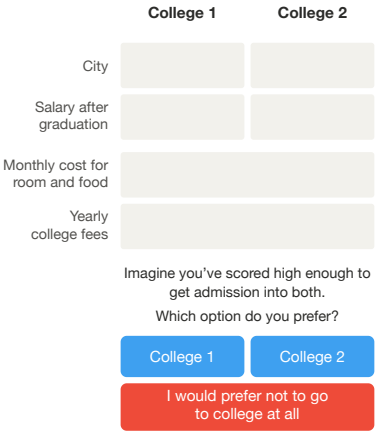
2. What does $\Delta NonPec_{id}$ consist of?
   Multidimensional, subjective $\Rightarrow$ elicit destination-specific beliefs about non-pecuniary outcomes

**Experiment II:** How do educational migration choices respond to information?

1. Access to information and choices are endogenous $\Rightarrow$ randomize contact with migrant peers

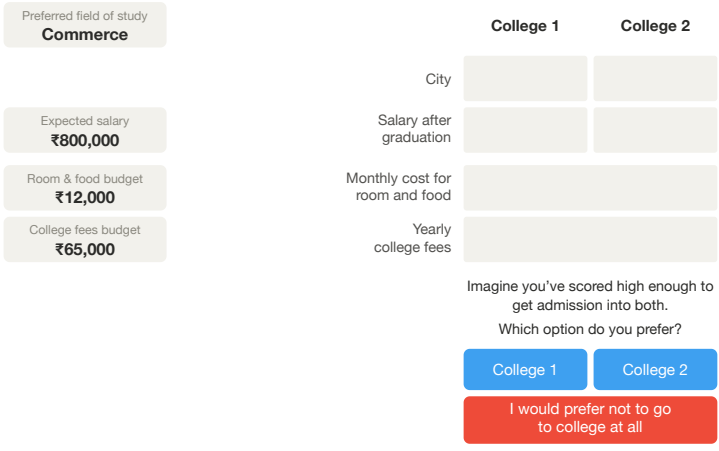
# Experiment I: Design

# Students choose between pairs of feasible, hypothetical colleges



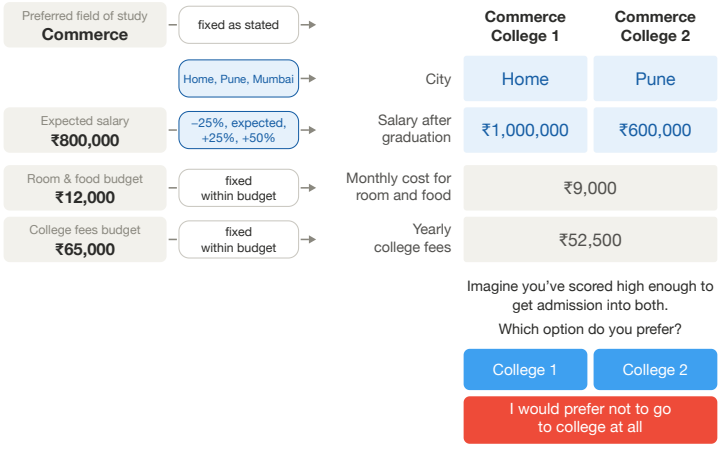

**Design:** Holding all pecuniary aspects constant, location of college captures $\Delta NonPec_{id}$

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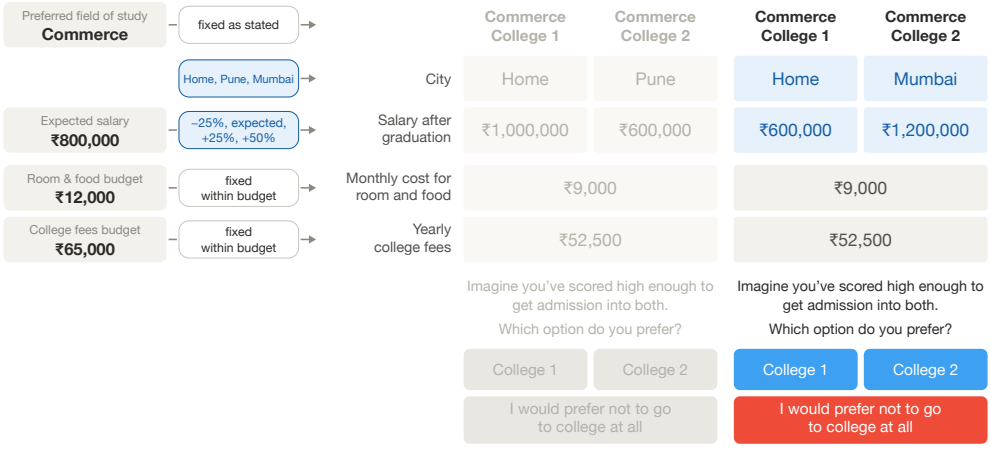

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| | Commerce College 1 | Commerce College 2 | Commerce College 1 | Commerce College 2 | **Commerce College 1** | **Commerce College 2** |
|---|---|---|---|---|---|---|
| City | Home | Pune | Home | Mumbai | Pune | Mumbai |
| Salary after graduation | ₹1,000,000 | ₹600,000 | ₹600,000 | ₹1,200,000 | ₹800,000 | ₹800,000 |
| Monthly cost for room and food | ₹9,000 | | ₹9,000 | | ₹9,000 | |
| Yearly college fees | ₹52,500 | | ₹52,500 | | ₹52,500 | |
| | Imagine you've scored high enough to get admission into both. Which option do you prefer? | | Imagine you've scored high enough to get admission into both. Which option do you prefer? | | Imagine you've scored high enough to get admission into both. Which option do you prefer? | |
| | College 1 | College 2 | College 1 | College 2 | College 1 | College 2 |
| | I would prefer not to go to college at all | | I would prefer not to go to college at all | | I would prefer not to go to college at all | |

**Design:** Holding all pecuniary aspects constant, location of college captures $\Delta NonPec_{id}$

# Empirical specification: disutility of migration in expected salary terms

**In choice task $t$, student $i$ chooses between college profiles and the outside option, indexed by $j \in D_{it}$**

$$U_{ijt} = \beta_0 \text{NoUniversity}_{ijt} + \boldsymbol{\delta}' \mathbf{D}_{ijt} + \boldsymbol{\lambda}' \mathbf{S}_{ijt} + \varepsilon_{ijt}, \qquad \varepsilon_{ijt} \overset{\text{i.i.d.}}{\sim} \text{type I extreme value}$$

$\delta_d =$ utility of destination $d$ relative to home

$\lambda_s =$ utility of salary level $s$ relative to expected salary

# Empirical specification: disutility of migration in expected salary terms

**In choice task $t$, student $i$ chooses between college profiles and the outside option, indexed by $j \in D_{it}$**

$$U_{ijt} = \beta_0 \text{NoUniversity}_{ijt} + \boldsymbol{\delta}' \mathbf{D}_{ijt} + \boldsymbol{\lambda}' \mathbf{S}_{ijt} + \varepsilon_{ijt}, \qquad \varepsilon_{ijt} \overset{\text{i.i.d.}}{\sim} \text{type I extreme value}$$

$\delta_d =$ utility of destination $d$ relative to home

$\lambda_s =$ utility of salary level $s$ relative to expected salary

**Utility from a 1% change in salary**

$$\widehat{\beta}_{\%\text{Salary}} \approx \frac{1}{3}\left(\frac{\widehat{\lambda}_{-25\%}}{-25} + \frac{\widehat{\lambda}_{+25\%}}{25} + \frac{\widehat{\lambda}_{+50\%}}{50}\right)$$

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**Utility from migrating to $d$, expressed in expected-salary terms**

$$\widehat{\text{MRS}}_{d,\%\text{Salary}} \approx -\frac{\widehat{\delta}_d}{\widehat{\beta}_{\%\text{Salary}}}, \qquad d \in \mathcal{D}$$

# Complementary data: what students expect and know about destinations

Students answer questions about their expected outcomes and knowledge of **same destinations**

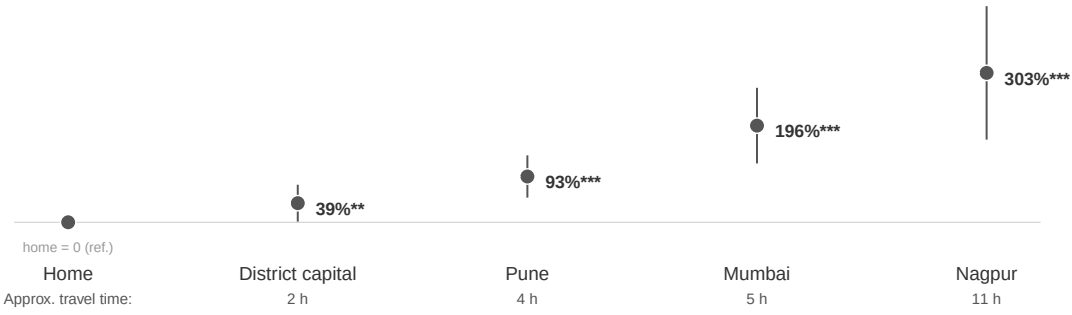
| | |
|---|---|
| **Pecuniary** | Expected earnings, chances of employment and costs<br>*"Think about a student who recently graduated from a [field] college in [destination] and is applying for jobs. What do you think is the typical yearly salary [she] can earn in [destination]?"* |
| **Non-pecuniary** | Emotional, social and practical life<br>*"How safe would you feel living and studying in [destination]?"* |
| **Knowledge** | About destinations, colleges, and contacts<br>*"How much do you know about studying and student life in [destination]?"* |

# Findings I

# Average student requires more than double the salary to leave home

**Migration costs (% increase in salary required to migrate to destination)**

Baseline and midline control data pooled, joint salary coefficient. Robust to attention checks.



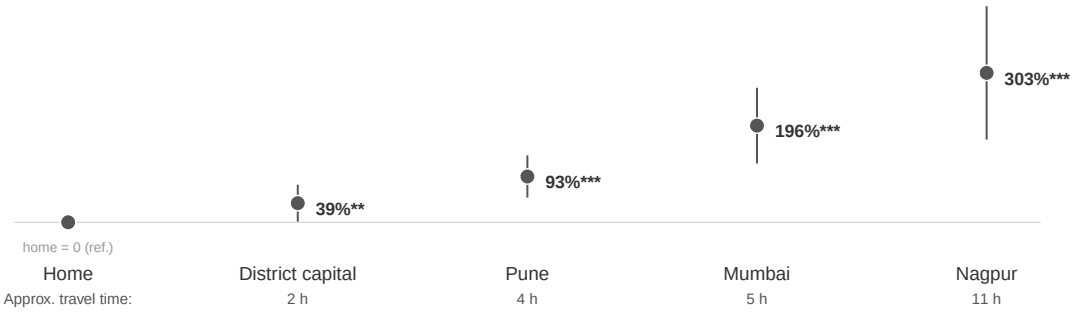

- Average migration costs of leaving home: 129% of expected salaries, holding all else constant

In USD | In USD, by gender | vs gains, by gender | Employment numeraire

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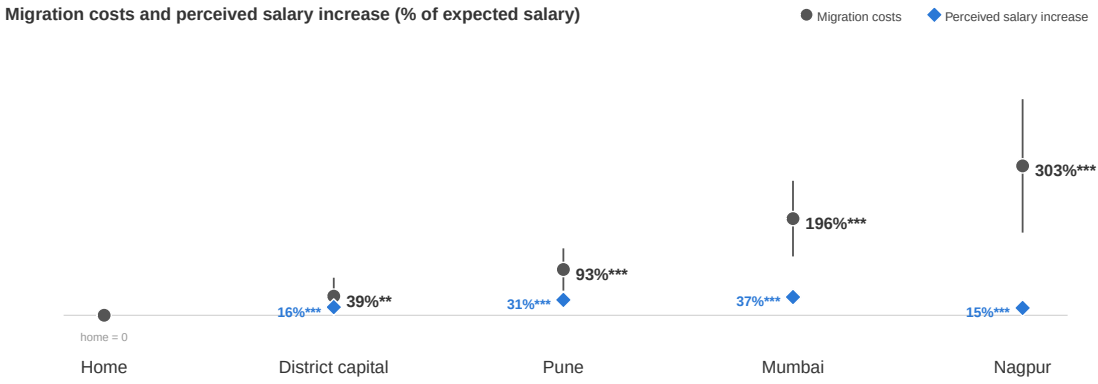
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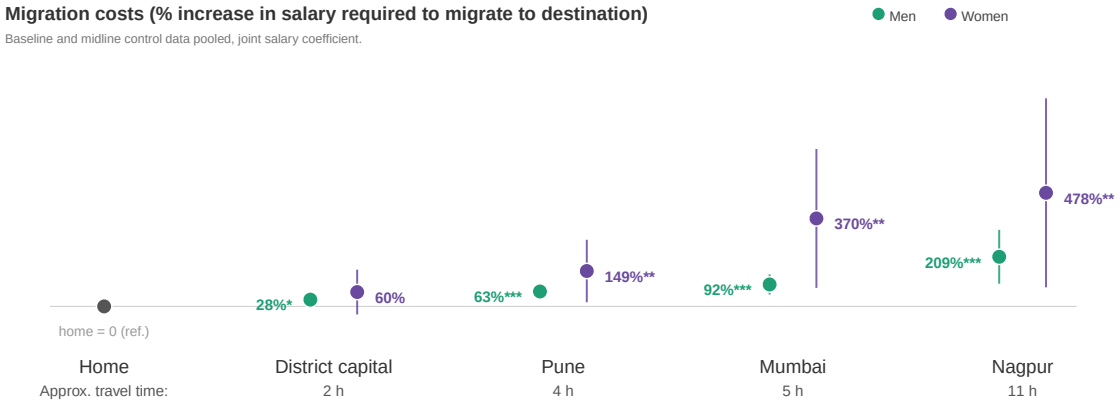
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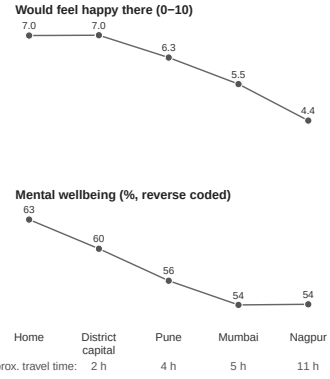



- Average migration costs of leaving home: 129% of expected salaries, holding all else constant
- Women require 3x men: weaker taste for higher salaries

In USD | In USD, by gender | vs gains, by gender | Employment numeraire

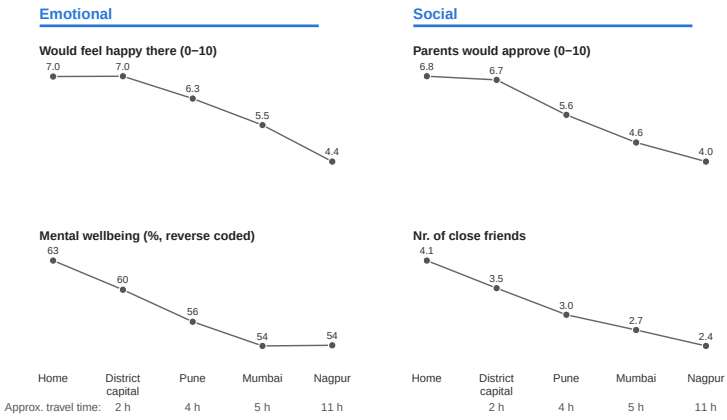
# Students expect worse non-pecuniary outcomes away from home

**Emotional**




By gender | As SD | As SD, by gender

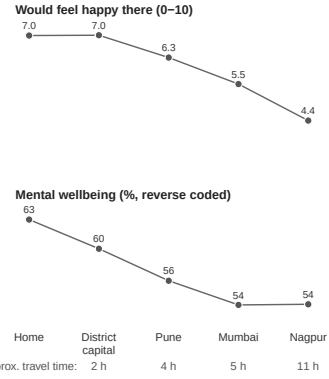
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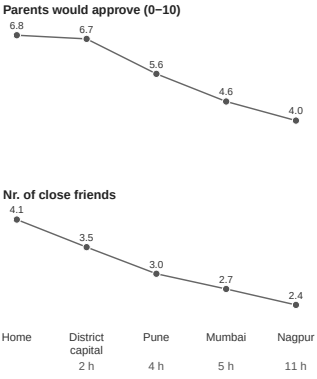

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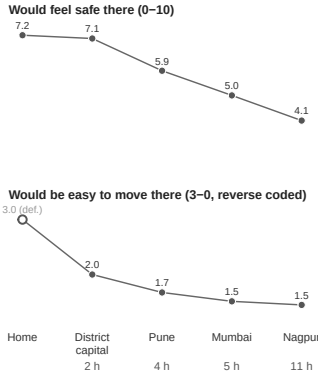
## Emotional




## Social

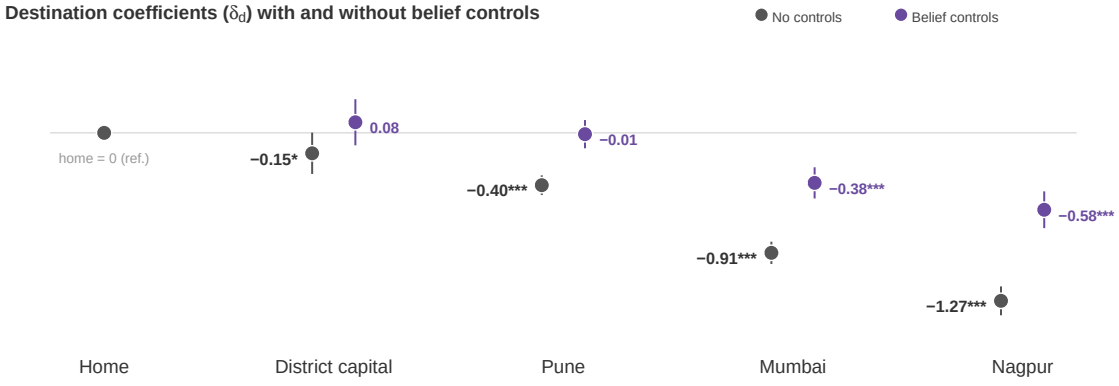



## Practical




By gender | As SD | As SD, by gender

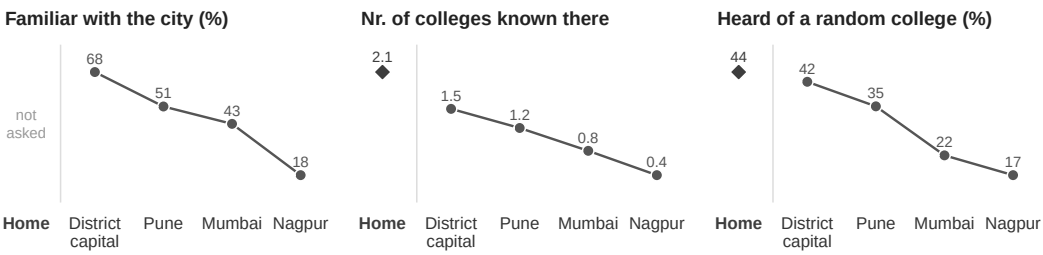
# Beliefs account for a substantial portion of migration costs




- Beliefs absorb 67% of the migration costs of leaving home — 12% for men, 98% for women
  (formal decomposition: 61% / 42% / 74%)

# Low awareness of college options and admissions processes

## Knowledge of the destination




- 29% lack college advice sources, only 4 schools (12% of students) have a college counselor; 30% have not considered any colleges at baseline

Information about college admissions | Information about destinations

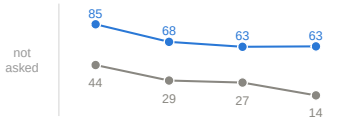
# A contact in a destination closes the gap between far and near cities

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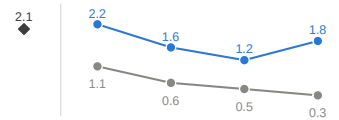
## Knowledge of the destination

Knows someone there
Doesn't know anyone

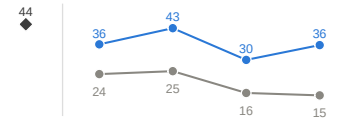
### Familiar with the city (%)




### Nr. of colleges known there

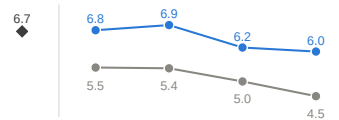



### Heard of a randomly drawn college (%)

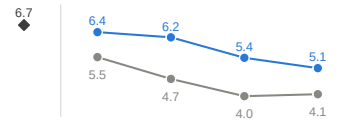



## Non−pecuniary beliefs about the destination

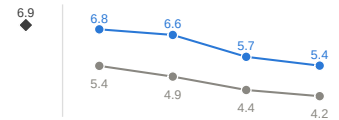
### Would feel happy there (0−10)




### Parents would approve (0−10)




### Would feel safe there (0−10)




Who are the contacts

# Summary: Experiment I

## What prevents students from migrating for better educational opportunities?

- Students expect higher earnings away from home and plan to stay home anyway:
  - → non-pecuniary migration costs 100%+ of expected earnings, 3x larger for women
  - → beliefs about safety, family and wellbeing accounting for a large share
- Existing social networks can transmit information and improve beliefs about destinations, but most do not have such a contact

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**Next**

How would students' beliefs and college choices change with access to a migrant peer?

# Experiment II: Design

# Intervention: phone mentorship with migrant college students

## Campus Trail program

- One-on-one phone sessions
- Pairs share gender/academic stream

## Mentors

- 51 college students in Pune and Mumbai
- Receive $10 per mentee and LoR



# What can a migrant peer change?

Mentors were equipped to address a large number of questions and concerns:

Mentee impressions

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- **Cost of living**

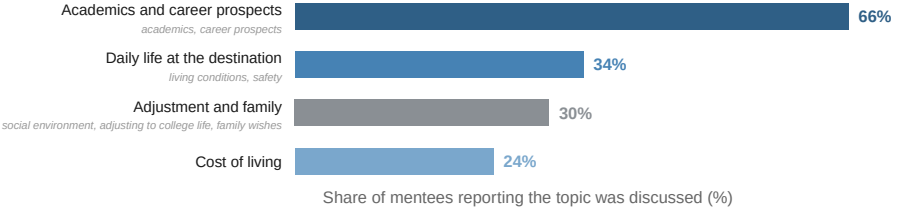
  *"I will first check his previous marks of SSC and HSC, because these marks matter a lot to get a government hostel for free."*

Mentee impressions

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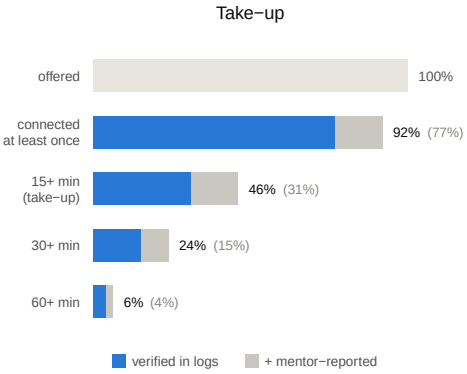
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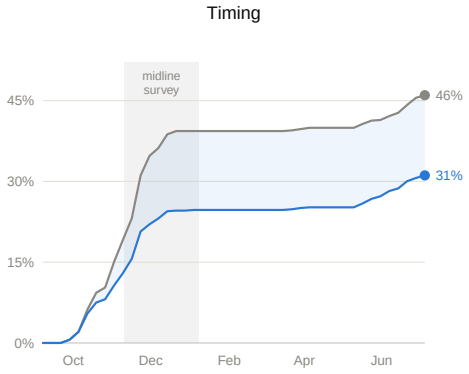
Mentees recall discussing a variety of topics:




Mentee impressions

# Take-up: engagement and timing




- **46%** of treatment students spoke with their mentors for at least 15 minutes (take-up)
- Among those who took up treatment, average call duration is appx. **40 minutes**

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**Intent to treat:** $Y_i = \alpha + \beta\, \text{CampusTrail}_i + \varepsilon_i$ robust standard errors

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- Applied
- Applied for financing
- Enrolled

**Intensive margin: migration**

- Applied away from home
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**Beliefs about destinations**

$$B_{id} = \alpha + \beta_1\,\text{CampusTrail}_i + \beta_2\,\text{Away}_d + \beta_3\,\text{CampusTrail}_i \times \text{Away}_d + \mathbf{X}'_{id}\gamma + \varepsilon_{id}$$

- $B_{id}$: belief indices for the district capital and Pune, normalized to control home
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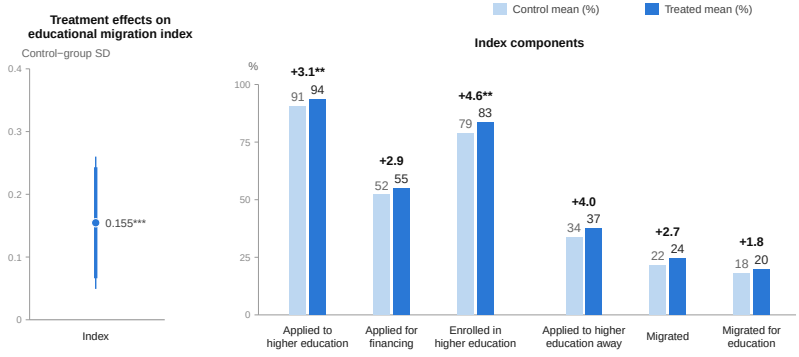
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- $B_{id}$: belief indices for the district capital and Pune, normalized to control home
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**Attrition** 84% surveyed at endline (68% directly); 91% with tracking weights Bounds

# Findings II

# Mentorship increased educational migration

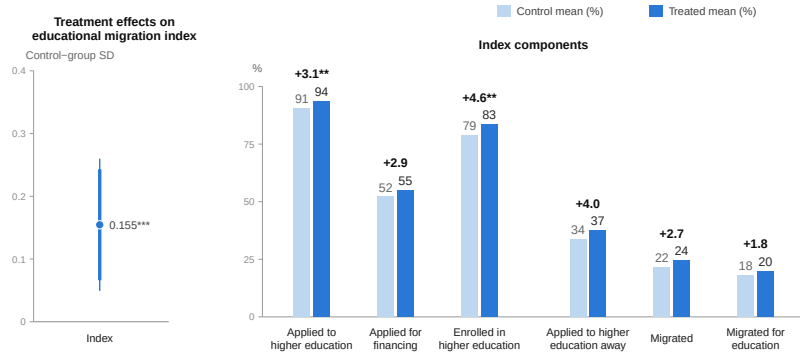



- Educational migration index: $+0.16$ SD

Outcomes are past or planned within two months: index restricted to past outcomes only 0.20 SD.

Index robustness Past only

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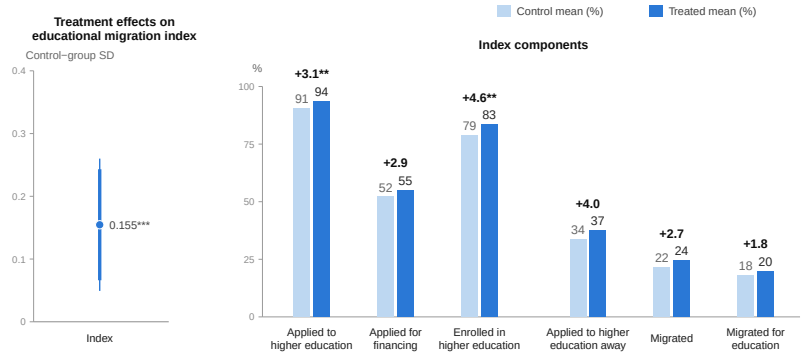



- Educational migration index: $+0.16$ SD
- Extensive margin: applications $+3.1$ pp, financing $+2.9$ pp, enrollment $+4.6$ pp

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- Educational migration index: $+0.16$ SD
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- Intensive margin: applications away $+4.0$ pp, migration $+2.7$ pp; concentrated in colleges

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# Mentorship induces short moves within the home district




- Applications and moves to district capital increases, from other parts of the district

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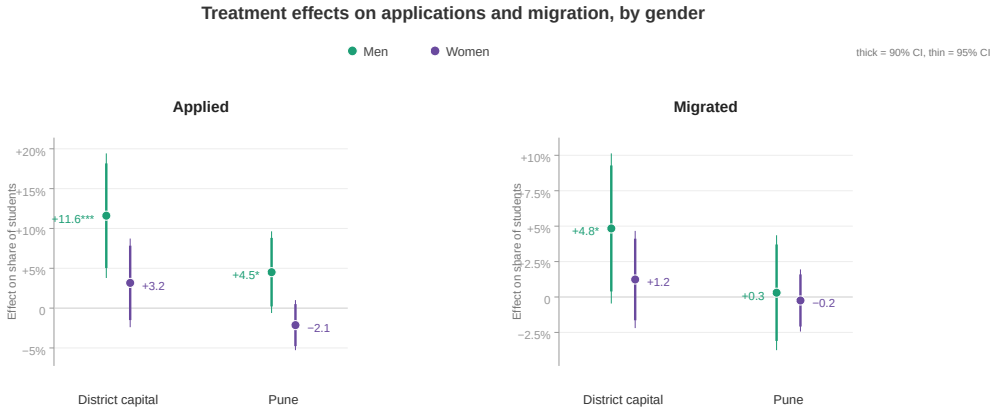
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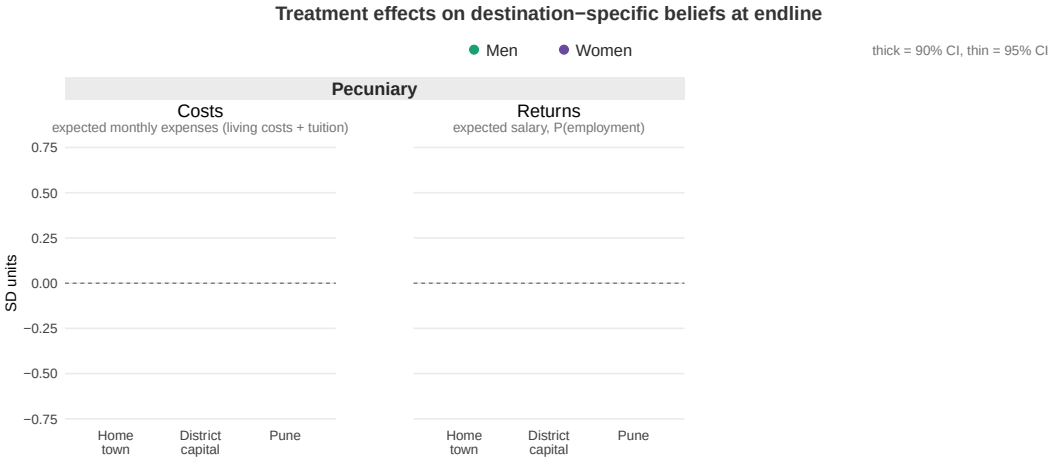

- Applications and moves to district capital increases, from other parts of the district
- No effect on applications or migration to the large metropolitan areas or to the mentor destinations
- Treatment effects entirely concentrated on students outside the district capital (0.21 SD)

ITT

# Men apply and move away; women don't



# Treatment effects on destination-specific beliefs mirror applications



# Treatment effects on destination-specific beliefs mirror applications



# Treatment effects on destination-specific beliefs mirror applications



# Treatment effects on destination-specific beliefs mirror applications



# Treatment effects on destination-specific beliefs mirror applications



# Summary: Experiment II

## How do beliefs and migration outcomes respond to new information?

- A light touch intervention, conversation with migrant peers can change choices:
  - → applications away $+4.0$ pp, migration $+2.7$ pp, concentrated in short moves
- However, it does not shrink migration costs where the largest earnings gains are:
  - → Beliefs about the far destinations moved **negatively**, if anything
- No detectable effects on any other pre-specified channel Other mechanisms

# Conclusion

# Conclusion

- Educational migration can be key to **reducing spatial inequality** (Ghose, 2026)
- Scholarships or information on living costs can help students make **the shortest moves**
- However, students expect **much larger practical, social, and emotional costs** of moving far
- If these beliefs are broadly accurate — and worsening upon contact with mentors suggests they are — **encouraging educational migration at scale likely means lowering the non-pecuniary costs at destinations,** not correcting misperceptions
  → e.g., safe and affordable student housing, transport, moving and travel assistance

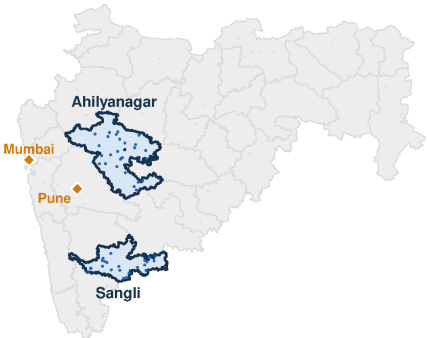
Thank you!

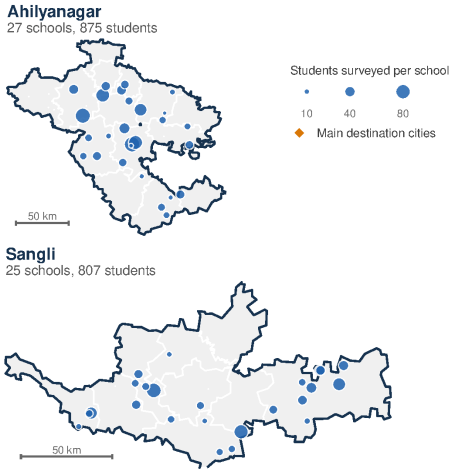
Contact:

merve.demirel@iies.su.se

# Backup: setting

# Surveyed schools in Ahilyanagar and Sangli

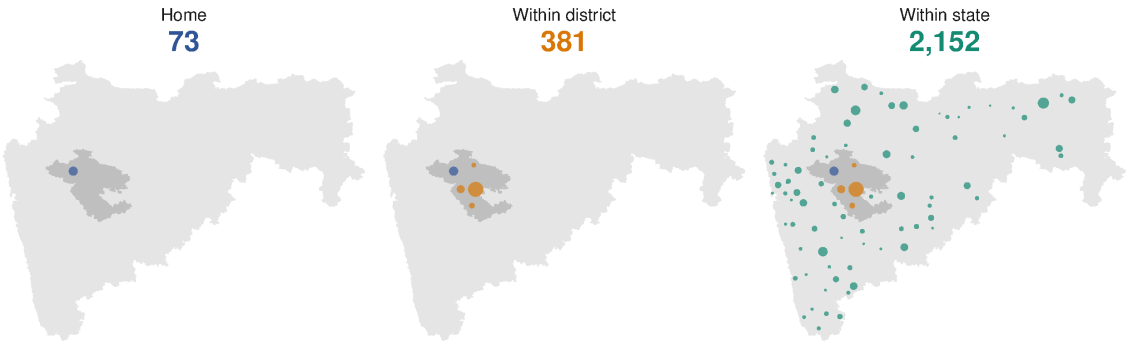



Back

# Accessible seats: 25th-percentile applicant

**Number of accessible spots per qualifying candidate, 25th–percentile engineering applicant**

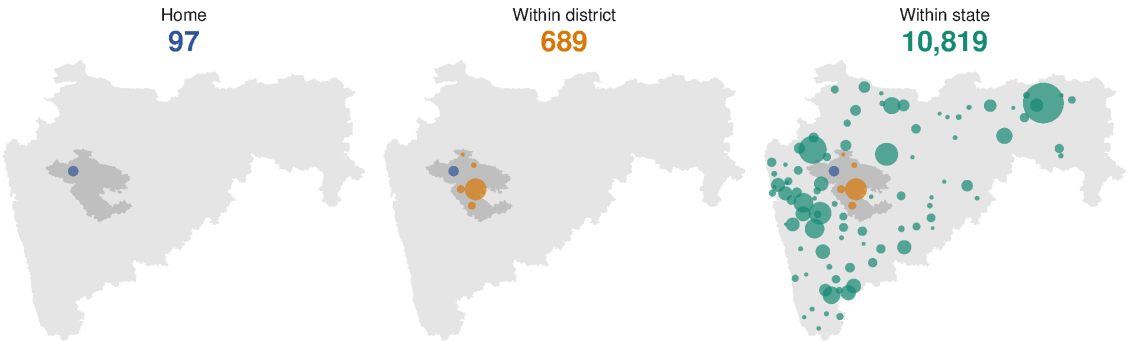



Source: administrative admissions records of the State Common Entrance Test Cell, Government of Maharashtra.

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# Accessible seats: 75th-percentile applicant

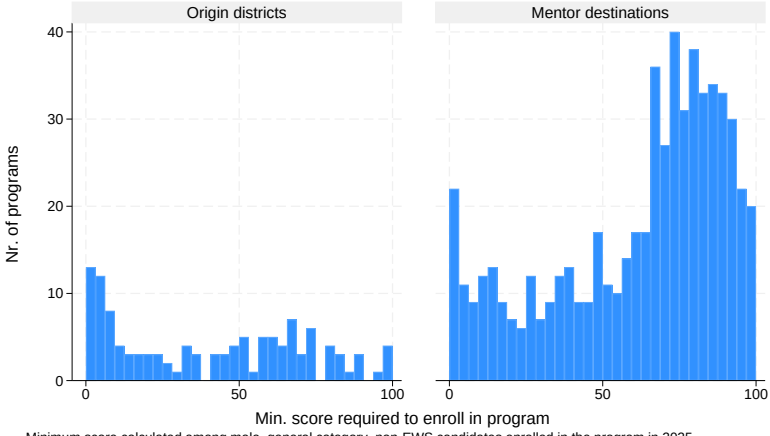
**Number of accessible spots per qualifying candidate, 75th–percentile engineering applicant**




Source: administrative admissions records of the State Common Entrance Test Cell, Government of Maharashtra.

Back

# Origin districts have fewer, less competitive programs




Minimum score calculated among male, general category, non-EWS candidates enrolled in the program in 2025.

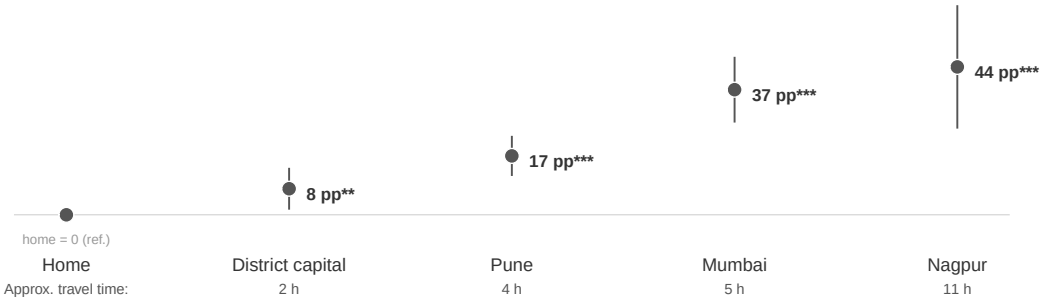
Back

# Backup: baseline facts

# Migration costs in employment terms

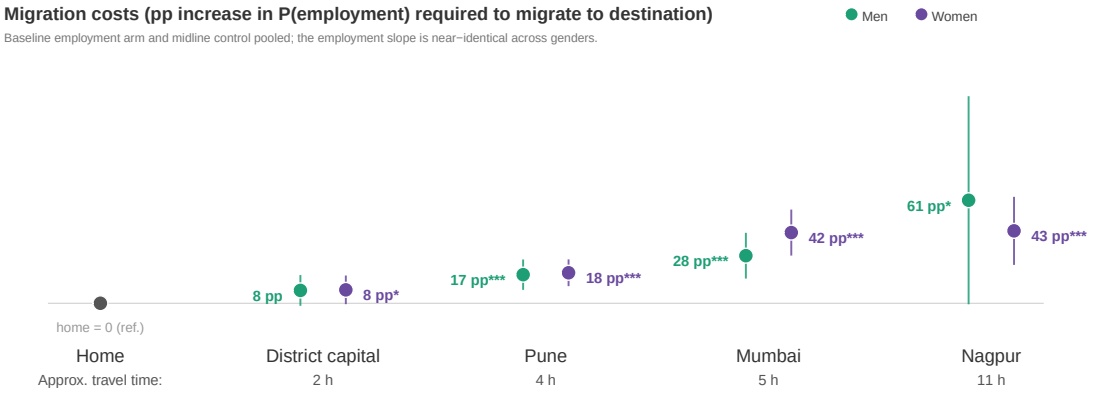
**Migration costs (pp increase in P(employment) required to migrate to destination)**

Baseline employment arm and midline control pooled as in the salary ladder; the same midline sets identify the district−capital coefficient.



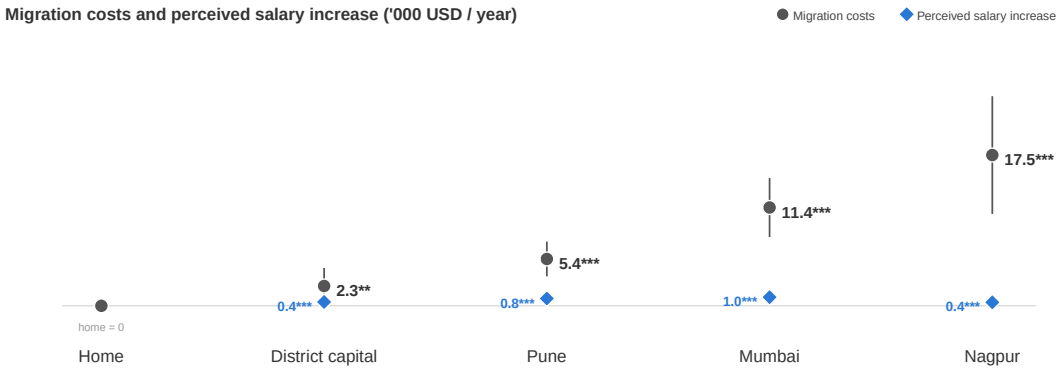

Back

# Migration costs in employment terms




Back

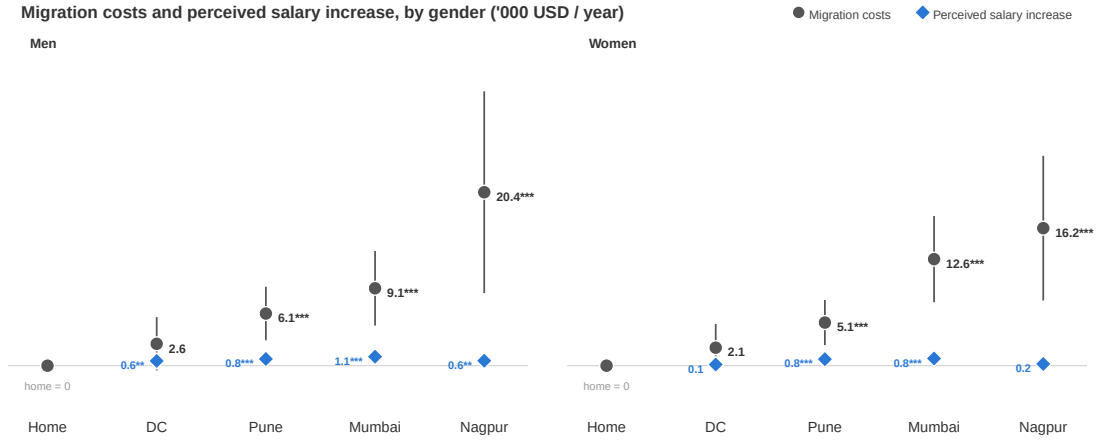
# Migration costs against perceived salary gains




By gender

Back

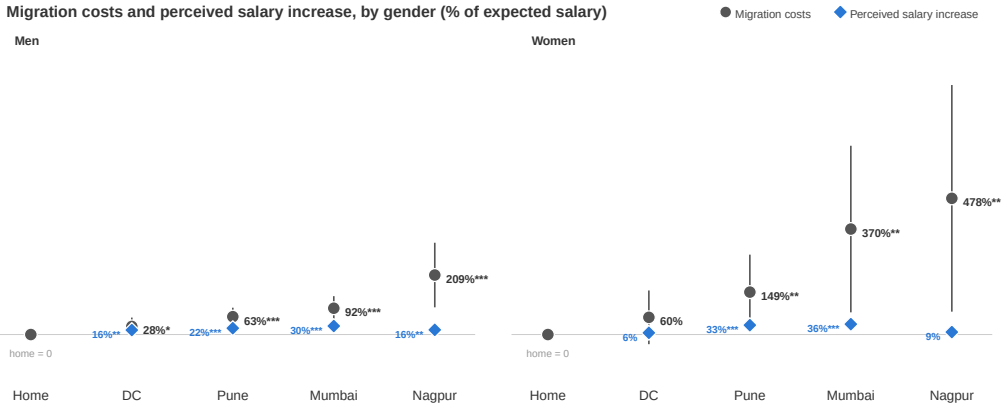
# Migration costs against perceived salary gains, by gender




Pooled

Back

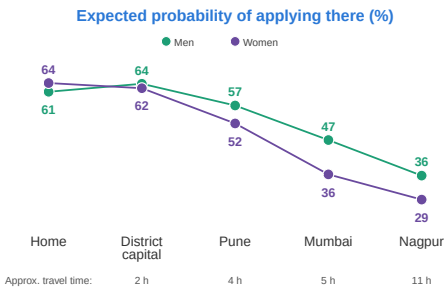
# Migration costs against perceived salary gains, by gender

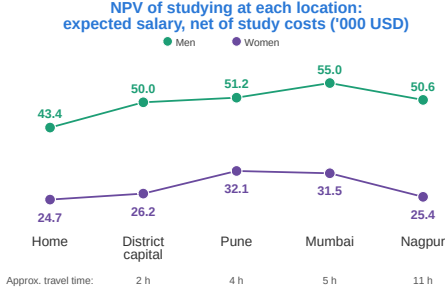



- Women expect much lower salaries, and therefore require higher increases to leave

Back

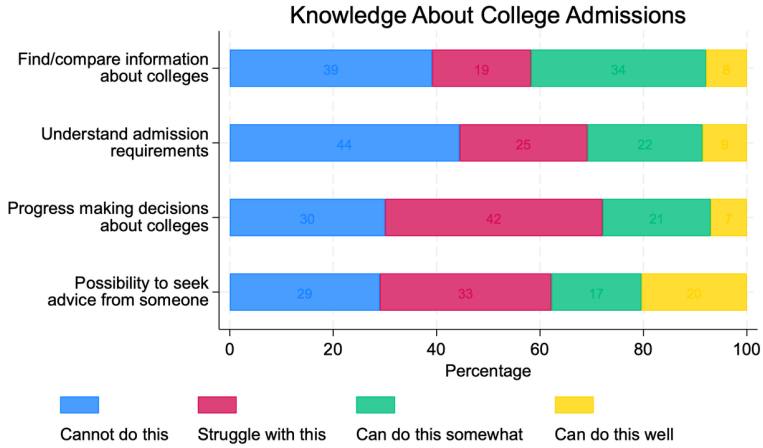
# Stated choice and expected returns, by gender




Back

# Generally low levels of knowledge

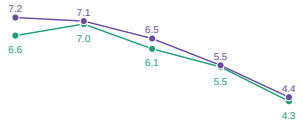



Back

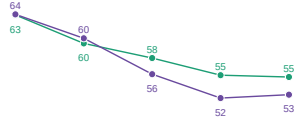
# By gender: worse non-pecuniary outcomes and weakened social ties

## Emotional

### Would feel happy there (0–10)

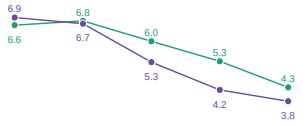



### Mental wellbeing (%, reverse coded)

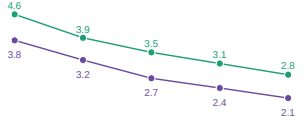



## Social

### Parents would approve (0–10)

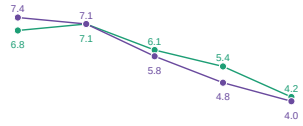



### Nr. of close friends

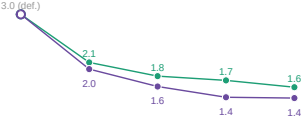



## Practical

### Would feel safe there (0–10)




### Would be easy to move there (3–0, reverse coded)




Home | District capital | Pune | Mumbai | Nagpur

Approx. travel time: 2 h | 4 h | 5 h | 11 h

Men | Women

Back

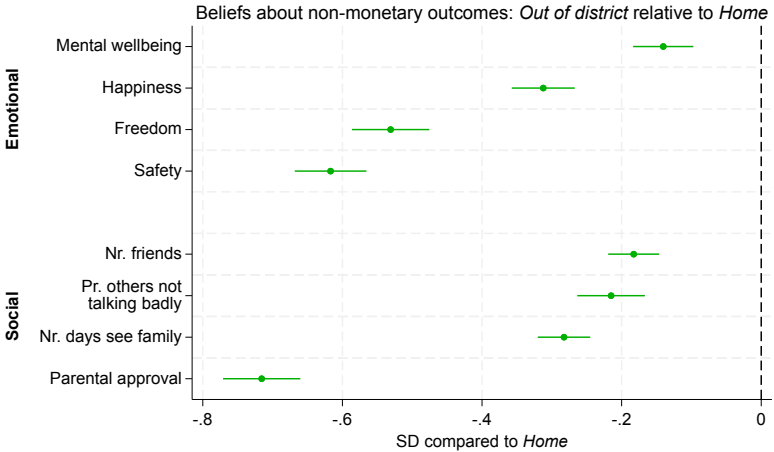
# Who students know at each destination

| | District capital | Pune | Mumbai | Nagpur |
|---|---|---|---|---|
| *A. Share of students who know someone there (%)* | | | | |
| Any contact | 57 | 57 | 45 | 9 |
| Relatives | 41 | 44 | 36 | 5 |
| Family friends | 9 | 11 | 7 | 2 |
| Friends | 10 | 11 | 7 | 2 |
| Other acquaintances | 6 | 6 | 3 | – |
| Student–destination pairs | 225 | 1,806 | 1,713 | 848 |
| *B. Among students who know someone there: who (%)* | | | | |
| Relatives | 73 | 77 | 80 | 62 |
| Family friends | 16 | 19 | 15 | 19 |
| Friends | 18 | 19 | 15 | 29 |
| Other acquaintances | 10 | 11 | 7 | – |
| Pairs with a contact | 128 | 1,037 | 763 | 73 |

Unit: student–destination pair (baseline, all students; midline, control group). Contact types overlap: a student can report several. "Other acquaintances" was asked at midline only; the district capital was asked about at midline only, Nagpur at baseline only.

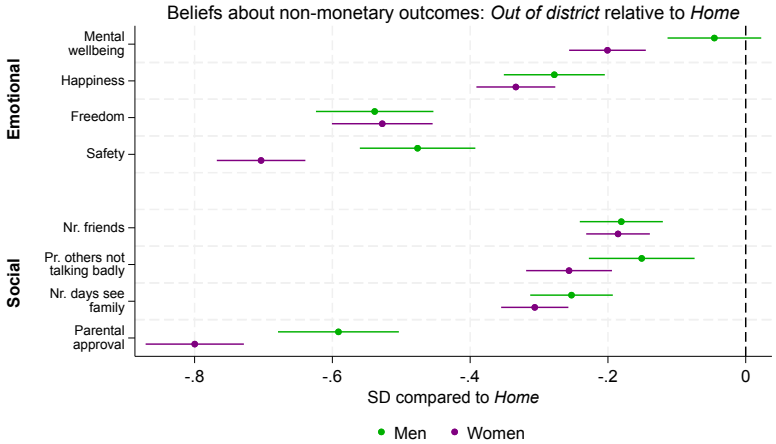
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# Beliefs about away destinations, in SD units



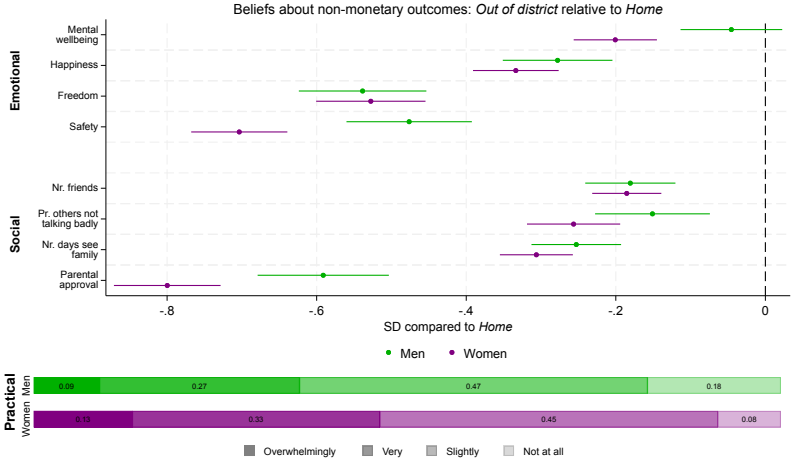

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# By gender: worse non-pecuniary outcomes and weakened social ties




Back

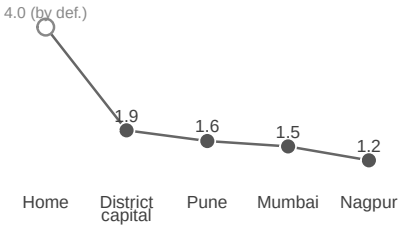
# By gender: worse non-pecuniary outcomes and weakened social ties



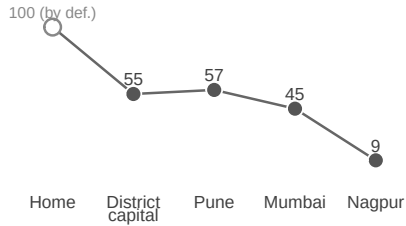

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# Knowledge of away destinations

**Knows the city (1 = nothing, 4 = really well)**



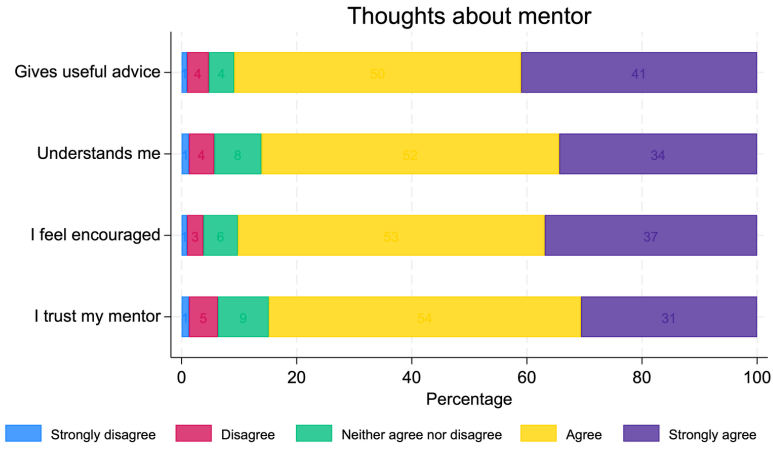

**Has any contacts there (%)**




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# Backup: intervention and data

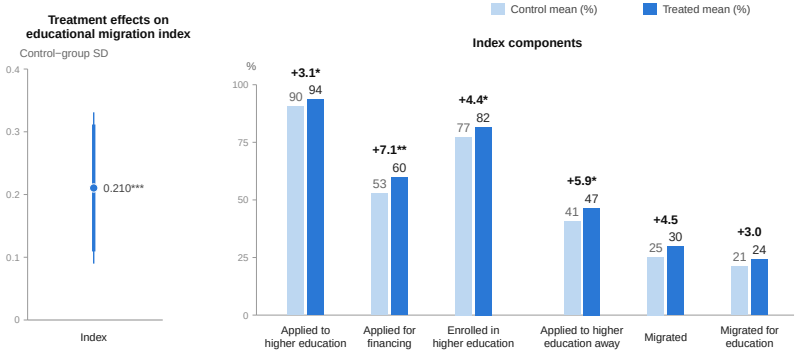
# Students report positive impressions of interactions with mentor




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# Backup: results

# Treatment effects are driven by students outside the district capital

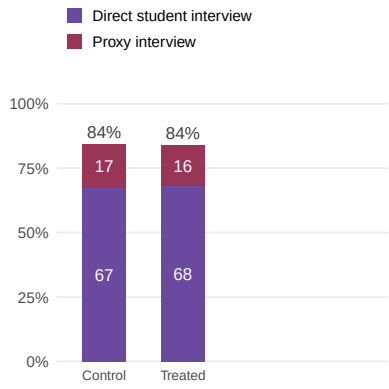



- Educational migration index: $+0.21$ SD
- Extensive margin: applications $+3.1$ pp, financing $+7.1$ pp, enrollment $+4.4$ pp
- Intensive margin: applications away $+5.9$ pp, migration $+4.5$ pp

Excludes the quarter of students residing in a district capital.

# Attrition

Back Endline data collected from students when available, otherwise proxy (a parent)



# Attrition

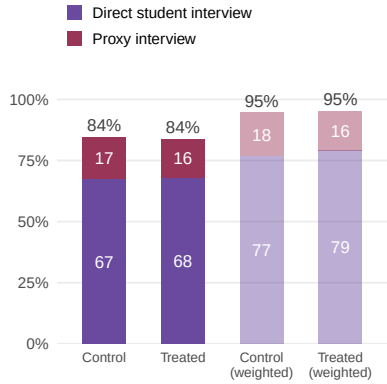
Back Endline data collected from students when available, otherwise proxy (a parent)

## Selective

Baseline traits predict who is surveyed at endline

Intensive-tracking weights, double-sampling worst-case and sensitivity calibrated to the phone-vs-intensive gap bounds.

Bounds



# Attrition

Back Endline data collected from students when available, otherwise proxy (a parent)

## Selective

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Intensive-tracking weights, double-sampling worst-case and sensitivity calibrated to the phone-vs-intensive gap bounds.

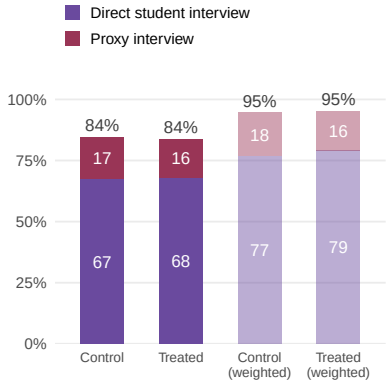
Bounds

## Differential

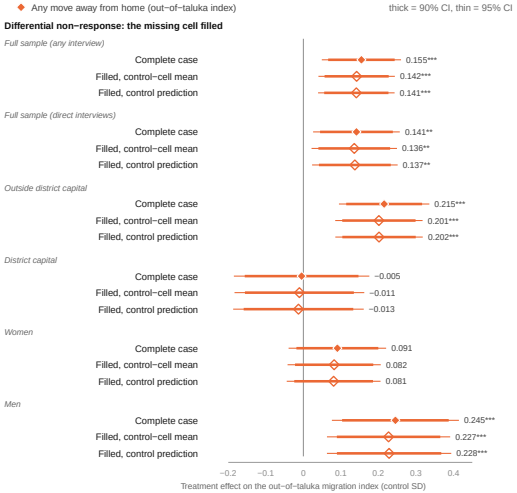
None by treatment status, but treated students who don't want to move at baseline are differentially missing

Would inflate observed effect; findings robust to imputation and bounding exercises.

Imputation



# Differential attrition

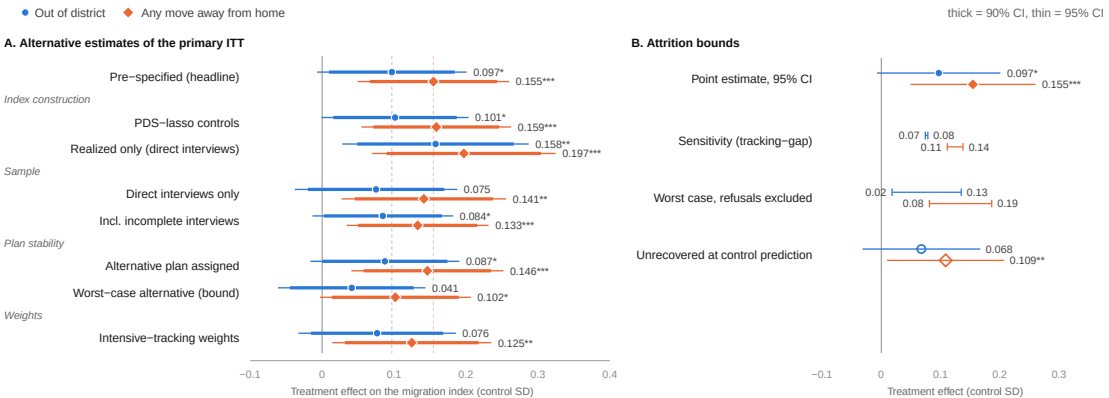



Excess non-response within baseline-migration intention cells filled at control-respondent cell means or predicted values; 90/95% bootstrap CIs.

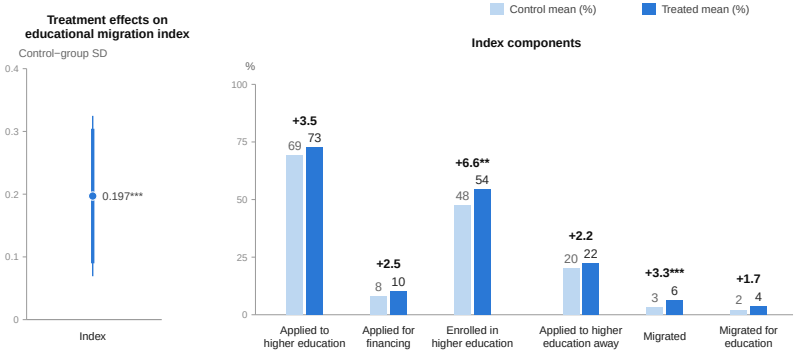
Back to attrition

# Robustness: alternative estimates and attrition bounds

Back to ITT Back to attrition



# Past only: realized outcomes




Same construction as the main figure, with each component restricted to actions already taken at the interview; student interviews.

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# Other mechanisms: no detectable effects

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- **Destination knowledge:** the city, colleges there, contacts and college students known
- **Admissions information:** requirements, comparing colleges, financing, options considered
- **Effort:** college search, study hours, HSC test scores
- **Support and encouragement:** support in deciding, preparedness to move
- **Traits:** self-efficacy, gender attitudes, aspirations

Information and search over time

# Search and information rise in both arms from baseline to midline

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